KICKPAD

SMART CONTRACT AUDIT

ZOKYO.

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PASS

Zokyo's Security Team has concluded that this smart contract passes security qualifications to be listed on digital asset exchanges.

TECHNICAL SUMMARY

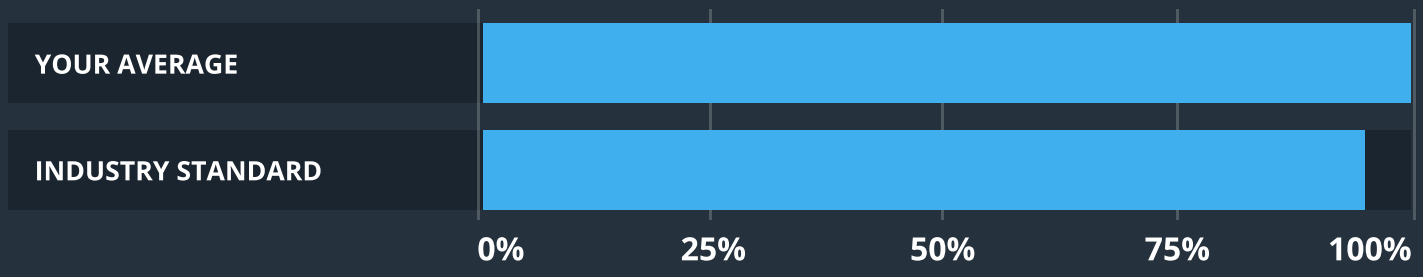
This document outlines the overall security of the Kick.Finance smart contracts, evaluated by Zokyo's Blockchain Security team.

The scope of this audit was to analyze and document the Kick.Finance smart contract codebase for quality, security, and correctness.

Contract Status

There was one critical issue found during the audit, which was successfully resolved.

Testable Code



Testable code is 100% which is above the industry standard of 95%.

It should be noted that this audit is not an endorsement of the reliability or effectiveness of the contract, rather limited to an assessment of the logic and implementation. In order to ensure a secure contract that's able to withstand the Ethereum network's fast-paced and rapidly changing environment, we at Zokyo recommend that the Bondly team put in place a bug bounty program to encourage further and active analysis of the smart contract.

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AUDITING STRATEGY AND TECHNIQUES APPLIED

The Smart contract's source code was taken from the following github repository – https://github.com/bnbguy1/Kick/.
Commit id – edd95d8dd70a5bd24ecd9e69477103ac6bf190f5.

Zokyo team audited only one file - **KickPad.sol Solidity**.

Throughout the review process, care was taken to ensure that the token contract:

- Implements and adheres to existing Token standards appropriately and effectively;
- Documentation and code comments match logic and behavior;
- Distributes tokens in a manner that matches calculations;
- Follows best practices in efficient use of gas, without unnecessary waste;
- Uses methods safe from reentrance attacks;
- Is not affected by the latest vulnerabilities;
- Whether the code meets best practices in code readability, etc.

Zokyo's Security Team has followed best practices and industry-standard techniques to verify the implementation of Kick.Finance smart contracts. To do so, the code is reviewed line-by-line by our smart contract developers, documenting any issues as they are discovered. In summary, our strategies consist largely of manual collaboration between multiple team members at each stage of the review:

1	Due diligence in assessing the overall code quality of the codebase.	3	Testing contract logic against common and uncommon attack vectors.
2	Cross-comparison with other, similar smart contracts by industry leaders.	4	Thorough, manual review of the codebase, line-by-line.

EXECUTIVE SUMMARY

There was one critical issue found during the audit, which was successfully resolved. All the mentioned findings may have an effect only in case of specific conditions performed by the contract owner.

Contracts are well written and structured. The findings during the audit have no impact on contract performance or security, so it is fully production-ready.

STRUCTURE AND ORGANIZATION OF DOCUMENT

For ease of navigation, sections are arranged from most critical to least critical. Issues are tagged “Resolved” or “Unresolved” depending on whether they have been fixed or addressed. Furthermore, the severity of each issue is written as assessed by the risk of exploitation or other unexpected or otherwise unsafe behavior:

Critical

The issue affects the ability of the contract to compile or operate in a significant way.

High

The issue affects the ability of the contract to compile or operate in a significant way.

Medium

The issue affects the ability of the contract to operate in a way that doesn’t significantly hinder its behavior.

Low

The issue has minimal impact on the contract’s ability to operate.

Informational

The issue has no impact on the contract’s ability to operate.

MANUAL REVIEW

CRITICAL | RESOLVED

Token contract has total supply equal to 203768315, but Whitepaper indicates total supply as 203589744.

Recommendation:
Change total supply in Token contract to number indicated in Whitepaper.

FIX:
The token number in whitepaper is incorrect. Kick.Finance team have updated the whitepaper to show the correct token amounts. The amount of tokens is 203768315. This was a typo in the Whitepaper.

HIGH | RESOLVED

Audited repo does not have vesting & tokens distributions contract or functionality.

Recommendation:
Allocate and distribute tokens for vesting according to Whitepaper.

FIX:
Kick.Finance team has created and tested lockup/vesting contracts for all the token categories. The vesting contracts we have correlate to each and every token lockup schedule. These lockup contracts are separate from the token contracts and will be put in place, with locked tokens, prior to launch.

MEDIUM | UNRESOLVED

Solidity compiler pragma is not pointing to exact version.

Recommendation:
Define specific Solidity compiler version in pragma keyword.

INFORMATIONAL | UNRESOLVED

Smart contract is not covered by NatSpec annotations.

Recommendation:
Cover by NatSpec all Contract methods.

INFORMATIONAL | UNRESOLVED

DRY, Solidity file has hard coded SafeMath & ERC-20 interface.

Recommendation:
Use openzeppelin contracts (or similar) as NPM dependency and utilize it inside of contracts.

CODE COVERAGE AND TEST RESULTS FOR ALL FILES

Tests written by Zokyo Secured team

As part of our work assisting Kick.Finance in verifying the correctness of their contract code, our team was responsible for writing integration tests using the Truffle testing framework.

Tests were based on the functionality of the code, as well as a review of the Kick.Finance contract requirements (for Audit) for details about issuance amounts and how the system handles these.

```
Contract: KickPad contract
  ✓ should set deployer correctly (65ms)
  ✓ should transfer total supply to deployer correctly (175ms)
  ✓ should transfer tokens correctly (634ms)
  ✓ shouldn't transfer tokens to zero address (120ms)
  ✓ shouldn't transfer too big amounts of tokens (129ms)
  ✓ should increaseApproval tokens correctly (336ms)
  ✓ should decreaseApproval tokens correctly (361ms)
  ✓ should decreaseApproval tokens correctly when subtracted value exceeds allowed value (331ms)
  ✓ should transferFrom tokens correctly (432ms)
  ✓ shouldn't transferFrom to zero address (239ms)
  ✓ shouldn't transferFrom if owner has not enough  tokens (248ms)
  ✓ shouldn't transferFrom if allowed amount of tokens is not enough (228ms)

12 passing (6s)
```

FILE	% STMTS	% BRANCH	% FUNCS	% LINES	UNCOVERED LINES
contracts/	100.00	100.00	100.00	100.00	
KickPad.sol	100.00	100.00	100.00	100.00	
All files	**100.00**	**100.00**	**100.00**	**100.00**	

We are grateful to have been given the opportunity to work with the Kick.Finance team.

The statements made in this document should not be interpreted as investment or legal advice, nor should its authors be held accountable for decisions made based on them.

Zokyo's Security Team recommends that the Kick.Finance team put in place a bug bounty program to encourage further analysis of the smart contract by third parties.

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